

Student Outcome & Outlier Analysis Report

Based on Provided Raw Database Context

1. Overview of Data Constraints

The provided database context contains demographic and enrollment data (Student ID, Gender, School Name, Grade, Academic Year, District, and Category). The dataset **does not** contain explicit fields related to “training,” “performance metrics,” or qualitative assessment scores.

Therefore, to address the query regarding students who “benefitted greatly” and those “left behind,” this report relies strictly on the category field as a proxy indicator:

- **Benefitted Greatly / Completed:** Students designated as Graduated.
- **Left Behind / Ongoing:** Students designated as Active.
- **Outliers:** Records with missing categorical status or non-standard grade levels.

2. Proxied Benefit Status

Based on the raw data sample provided, the vast majority of students are categorized as either Active or Graduated.

- **Graduated (Benefitted / Completed):** Represent the portion of the student body that has successfully finished their respective curriculum tier and progressed.
- **Active (In Process / Left Behind):** Represent students who are still moving through the system and have not yet reached graduation.

3. Outlier Analysis

Outliers are identified strictly from anomalous data entries in the raw dataset. These fall into two distinct groups: non-standard academic grades and missing categorical data.

Outlier Group A: Non-Standard Grade Level (Higher Education)

The primary focus of the dataset is secondary education (Grades 7 through 12). However, a distinct group of students is enrolled at the “Bachelor” grade level. These represent structural outliers in terms of expected age and academic progression within this dataset.

Student ID	Gender	School Name	Grade	Category
CLEC-S-0005865	Girl	Sotang Public Campus	Bachelor	Graduated
CLEC-S-0005839	Boy	Sotang Public Campus	Bachelor	Graduated
CLEC-S-0005697	Girl	Sotang Public Campus	Bachelor	Graduated
CLEC-S-0005681	Boy	Sotang Public Campus	Bachelor	Graduated
CLEC-S-0005690	Girl	Sotang Public Campus	Bachelor	Graduated
CLEC-S-0005835	Girl	Sotang Public Campus	Bachelor	Graduated

Outlier Group B: Missing Category Data

A small subset of students lacks a category classification entirely (the field is recorded as empty ' ').

Because their status is null, it is impossible to classify them as either “benefitted” (Graduated) or “left behind” (Active), making them statistical outliers.

Student ID	School Name	Grade	Academic Year
CLEC-S-0002778	Janahit Secondary School	9	2025
CLEC-S-0002706	Shree Prakash Jyoti Secondary School	9	2025
CLEC-S-0002713	Shree Prakash Jyoti Secondary School	9	2025
CLEC-S-0002773	Janahit Secondary School	9	2025
CLEC-S-0002790	Janahit Secondary School	10	2025
CLEC-S-0002779	Janahit Secondary School	9	2025
CLEC-S-0002832	Shree Siddhakali Secondary School	11	2025
CLEC-S-0002848	Shree Siddhakali Secondary School	10	2025

4. Conclusion

Without explicit training score fields, completion (Graduated) and ongoing enrollment (Active) serve as the sole available metrics for academic success and retention. The outliers highlight a need to track non-secondary (Bachelor) students separately and to resolve missing values in the core categorization system to ensure accurate impact reporting.